

NanoCore Malware Intelligence Documentation:

1. Executive Summary:

NanoCore is a Remote Access Trojan (RAT) first discovered in 2013, widely used by cybercriminals for espionage, credential theft, and persistence. It has been sold on underground forums and used in campaigns targeting multiple industries worldwide. This page organizes NanoCore intelligence into a structured format for long-term tracking, campaign analysis, and detection.

2. Scope:

- **Malware family intelligence**: NanoCore RAT
- **IOC management**: hashes, domains, IPs, mutexes, file extensions
- **Campaign tracking**: phishing and malspam delivery
- **Threat actor profiling**: criminal groups leveraging RATs for espionage and theft
- **Timeline analysis**: evolution from initial release to current variants

3. Core Intelligence Module Outputs:

- **Malware Family Page**: NanoCore aliases, technical behavior, infection chain

- **IOC Knowledge Base**: hashes, domains, IPs, mutexes
- **CampaignTimeline**: phishing /malspam campaigns from 2013–present
- **Detection Library**: YARA rules, signatures, persistence artifacts

4. Malware Family Documentation:

- **Family Name & Aliases**: NanoCore RAT, NanoCore Trojan
- **Discovery History**: First observed in 2013; commercialized on underground forums
- **Purpose & Evolution**: Initially marketed as a customizable RAT; evolved with plugins for credential theft, surveillance, and persistence
- **Technical Characteristics**:
 - Written in .NET
 - Plugin-based architecture
 - Remote desktop control, keylogging, password theft, webcam/microphone access .
- **File Extensions & Packaging**: Often delivered via .exe payloads in phishing emails, sometimes packed in .zip or .rar archives .
- **Known Hashes/YARA**: Multiple signatures exist in AV databases; YARA rules focus on NanoCore's plugin loader and mutex patterns .
- **Associated Campaigns**: Used in phishing campaigns targeting finance, education, and government sectors
- **TTPs**:
 - **Initial Access**: Phishing (T1566)

- **Execution**: Command and Scripting Interpreter (T1059)
- **Persistence**: Registry modification, scheduled tasks
- **Exfiltration**: Data sent via C2 servers over HTTP/HTTPS
- **Network Indicators**: Known C2 domains and IPs linked to NanoCore infrastructure
- **Persistence & Privilege Escalation**: Registry run keys, scheduled tasks, UAC bypass attempts
- **Unique File System Patterns**: Custom mutex names, plugin DLLs dropped in temp folders
- **References**: MalwareBazaar samples, reports from Palo Alto Networks, Trend Micro

5. Campaign Tracking Example

Campaign Tracking Example

Campaign ID	Threat Actor	Activity Period (UTC)	Victim Country	Target Industry	TTPs	IOC Summary	Notes
CAMP-NC-001	Criminal group (NanoCore sellers)	2013–2016	US, Europe	Finance, Education	T1566, T1059	Domains, hashes, IPs	Early NanoCore campaigns via phishing emails
CAMP-NC-002	Unknown actors	2017–2020	Global	Government, Manufacturing	T1059, T1071	IPs, mutexes	NanoCore variants with plugin extensions

6. Ransomware Module (Not Applicable):

NanoCore is not ransomware; no extortion patterns observed.

7. Final Deliverables:

- *NanoCore malware family page*
- *Campaign timeline dashboard*
- *IOC knowledge base (domains, hashes, mutexes)*
- *YARA detection rules*
- *Research notes and evidence tables*